SZEMERÉDI'S THEOREM ALONG CANTOR SETS OF INTEGERS

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Abstract. Some comments on your paper.

1. Comments on your paper

Dear friends,

I recently came across your arXiv preprint, "Szemerédi's theorem along Cantor sets of integers," and found the results quite interesting.

While reading, I noticed that you could immediately strengthen one of your results based on the proofs you have already established (you are currently claiming something weaker than what you have actually proved!). Additionally, I spotted a few technical points—including the characterization of my own previous work—that I believe need correction.

I have put together a concise list of these points below for your consideration (section 2).

1.1. **The Furstenberg and Katznelson IP theorem is already a density statement.**

In the paper, it is framed that your result establishes a density version of the Furstenberg-Katznelson theorem. I wanted to point out that Furstenberg and Katznelson already established a density version, as it follows directly from their IP theorem (see [FK85, Theorem A]) via the Furstenberg correspondence principle.

For clarity, the Furstenberg-Katznelson IP theorem asserts the following (see [1, Theorem A]):

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Definition 1.1. Let $(T_i)_{i\in\mathbb{N}}$ be a countable sequence of commuting transformations. To every finite subset $\alpha = \{i_1, ..i_k\}$, we attach the transformation

$$T_\alpha = T_{i_1} \circ ... \circ T_{i_k}.$$

The corresponding family is called an IP-system of transformations.

Theorem 1.2 (Furstenberg-Katznelson)**.** *Let $T^{(1)}, ..., T^{(\ell)}$ be ℓ-commuting IP-systems of transformations on a probability space $(X, \mathcal{B}, \mu)$. If $A \in \mathcal{B}$ and $\mu(A) > 0$, then there exists a finite subset $\alpha \subseteq \mathbb{N}$ so that*

$$\mu(A \cap (T^{(1)}_\alpha)^{-1}A \cap ... \cap (T^{(\ell)}_\alpha)^{-1}A) > 0.$$

To see the density implication, we may take a subset $E \subseteq \mathbb{N}$ of positive upper Banach density. Via the Furstenberg correspondence principle, we can produce a system $(X, \mathcal{B}, \mu, T)$ and a subset A with $\mu(A) = d(E) > 0$.

Now, for any IP, choose a generating sequence $n_1, n_2, ...$ and then define the IP-systems $T^{(i)}$, generated by $T^{i\cdot n_1}, T^{i\cdot n_2}, ...$ The theorem of Furstenberg and Katzanelson implies that we can find some α so that

$$\mu(A \cap T^{-\boldsymbol{n}_\alpha}A \cap T^{-2\boldsymbol{n}_\alpha}A \cap ...T^{-\ell\boldsymbol{n}_\alpha}A) > 0$$

where $\boldsymbol{n}_\alpha = \sum_{i\in\alpha} n_i$. Via the correspondence principle this shows that $d(E \cap E - \boldsymbol{n}_\alpha \cap ... \cap E - \ell \cdot \boldsymbol{n}_\alpha) > 0$. Because of this, the classical Furstenberg-Katznelson IP-theorem naturally encompasses the density result.

1.2. **Results for IP-sets already imply results for Cantor sets.** In the paper, it is framed that your results extend the Furstenberg-Katznelson IP theorem, as well as my joint work with Bryna Kra (partially). For example, you write: "It seems very plausible that many of the results of Kra and Shalom can be extended to the Cantor sets."

I wanted to point out that the logical implication actually flows in the other direction. As you correctly note in your paper, every Cantor set in the integers contains an IP. Because of this subset relationship, proving the existence of an arithmetic progression whose common difference lies in an IP set is a mathematically stronger statement than finding one in a Cantor set. Namely, if a theorem guarantees that a common difference can be found within any given IP set, it trivially guarantees that it can be found within any Cantor set (since we can just locate the IP set contained within that Cantor set).

Therefore, both the Furstenberg-Katznelson results and my results with Bryna Kra already apply directly to Cantor sets.

1.3. **What is new about your paper.** Following the previous points, I want to emphasize that while it might not be entirely accurate to frame your result as a generalization of the IP theorems, the converse is not true either. Your result is indeed new; the distinction simply lies in where the novelty is found. To clarify this, it helps to break down the types of questions typically asked in ergodic Ramsey theory:

- (Original Existence): Do sets of positive density contain progressions? In many cases (especially for arithmetic progressions), this is well-established (e.g., Szemerédi's theorem).
- (Type of Progression): What type of progressions? For instance, arithmetic progressions with a common difference in an IP set? In a Cantor set? Here, the smaller the set of allowed differences, the harder the question becomes.
- (Density of Progressions): Do sets of positive density contain many progressions (i.e., a positive density of them)? Furstenberg was the first to establish this type of result, and Furstenberg and Katznelson established it for the IP question (which, by extension, covers the Cantor set question).
- (Quantitative Density): What is the optimal density? This was answered by Bergelson, Host, and Kra for general arithmetic progressions of length $k = 3, 4$. The analogous result with the restriction that the difference must belong to a given rational IP set was established in my work with Bryna (again for $k = 3, 4$).
- (Frequency): How many differences admit this many progressions? Let E be a subset of positive density, and let A be the set of "good differences," defined as:

  $$A = \{n \in \mathbb{N} : d(E \cap (E - n) \cap \cdots \cap (E - kn)) > 0\}.$$

  Furstenberg's original work showed that A is syndetic. Furstenberg and Katznelson's theorem generalized this, showing that the set of differences is an IP* set (and thus also syndetic). In my work with

Bryna, we showed that within any IP set, the set of good differences has positive lower relative density.

Here is where your new contribution shines: You showed that for every Cantor set, you can find a syndetic set of indices $n \in \mathbb{N}$ such that $k_n \in A$. (Note: your draft currently only mentions positive lower density, but your proof actually establishes syndeticity, which is stronger!). To be precise, your result is not a density version of the Furstenberg-Katznelson IP Szemerédi theorem, nor is it an extension of it. However: *Your paper provides new, valuable insight into the frequency of these "good differences" (those in A) that also reside within a Cantor set.* The question of frequency is deeply interesting, yet it is less frequently studied in the literature. In my opinion, reframing your introduction around this specific contribution would highlight the true novelty of your work.

2. A short list

Here I write a short list of my comments on your paper.

2.1. **The abstract.**

(1) Unfortunately, I do not think your result extends Furstenberg and Katznelson's IP theorem. First because Furstenberg and Katznelson theorem is already a density theorem, but also, since every cantor set contains an IP, this means that finding progressions with difference in an IP is harder than finding those differences in the larger Cantor sets.

(2) You could improve the last sentence in the abstract: "In particular, this implies that for any subset of integers A of positive upper Banach density, there is a set B of integers n of positive lower Banach density such that A contains an ℓ + 1-term progression,with step size k_n, where $n \in B$." It seems to follow from your result that B is syndetic!

2.2. **The body of the paper.**

(3) It may not be obvious that Theorem 1.2 is a corollary of Theorem 1.3 (I can not find a proof of Theorem 1.2 in the paper). Specifically, in Theorem 1.3, one first average over x and then average over n. On

the other hand, in Theorem 1.2 you average over $x, n \in [N]$ and take N to infinity (so you take the average along x and along n to infinity simultaneously). Generally speaking if $\lim_{n\to\infty} \lim_{m\to\infty} a_{m,n} = a$, it does not immediately imply that $\lim_{n\to\infty} a_{n,n} = a$, e.g.

$$a_{m,n} = \begin{cases} 0 & m \geq n \\ 1 & m < n \end{cases}.$$

References

[1] H. Furstenberg and Y. Katznelson. An ergodic Szemerédi theorem for IP-systems and combinatorial theory. *J. Analyse Math.*, 45:117–168, 1985.

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